

15.6

(a) 在 EXPER 中, 兩年的迴歸效應非常接近 (1987 年為 0.1270, 1988 年為 0.1265), 且均具有迴歸效應遞減的特徵, 1988 年南方地區對 wage 的負面影響較 1987 年大 ($|-0.2384| > |-0.2128|$), 工會成員對 wage 的正面影響在 1987 年比 1988 年還大 ($0.1445 > 0.1102$)。在進行 OLS 估計時不考慮個體未觀察到的異質性, 故假設 u_i 與任何自變數不相關。#

(b) Panel data 假設同一個體在不同時間點的資料具有相關性, 因為彼此共用同個 u_i 。#

(c) 除了 intercept, EXPER 的係數變化最為劇烈, 說明在 OLS 中, EXPER 嚴重受到 omitted variable (如個人能力) 的正向偏誤影響。#

(d) $H_0: \beta_{11} = \beta_{12}, \beta_{12} = \beta_{13}, \dots, \beta_{1,N-1} = \beta_{1N}$ $H_1: \text{the } \beta_{1i} \text{ are not all equal}$	(15.19)	$F = \frac{(SSE_R - SSE_U)/(N-1)}{SSE_U/(NT - N - K_S)} \quad (15.20)$
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$N_{total} = 1432$, 個體數 $N = 716$, $T = 2$, $k = 4$

$df_1 = N - 1 = 716 - 1 = 715$

$df_2 = NT - N - k = 1432 - 716 - 4 = 712$

critical value $F_{0.01}(715, 712) = 1.13$

$\because F = 11.68 > CR = 1.13 \therefore \text{reject } H_0 \text{ at } \alpha = 0.01$

表示個體未觀察到的異質性確實存在, 迴歸 model 不適合用 Pooled OLS, 應使用 Fixed effect model 估計較合理。#

(e) FE: 假設同一個體在不同年份之間的誤差項是完全獨立且為 homoskedasticity

$$\Rightarrow \text{Cov}(e_{it}, e_{is}) = 0, t \neq s$$

FE robust: 允許同一個體在不同時間之間的 error term 符合 serial correlation 與 heteroskedasticity, 但不同人之間要互相獨立

調整成 cluster-robust standard error 後, SOUTH 和 UNION 的標準誤有明顯增加 (尤其 SOUTH 的標準誤幾乎增加一倍), 代表資料中存在個體 serial correlation, 傳統 FE 假設 error term 無相關會高估 t 值, 進而做出過度樂觀的判斷。#

(f)

EXPER 的係數在 FE 和 RE 之間有最大的差異 (FE: 0.0575 $\rightarrow$ RE: 0.0986), 代表 EXPER 受到嚴重的 omitted variable bias (u_i) 影響

$H_0: \text{cov}(x_{it}, u_i) = 0$ vs $H_1: \text{cov}(x_{it}, u_i) \neq 0$

Hausman test:

$$t = \frac{b_{FE} - b_{RE}}{\sqrt{[se(b_{FE})]^2 - [se(b_{RE})]^2}}$$

$$\Rightarrow t^* = \frac{0.0575 - 0.0986}{\sqrt{0.033^2 - 0.022^2}} = -1.671$$

$$RR: \{t \mid |t| \geq t_{2,0.25}(712) = 1.96\}$$

$\therefore t^* \notin RR \quad \therefore$ Do not reject H_0 at $\alpha = 0.05$

表示無法拒絕 $\text{cov}(x_{it}, u_i) = 0$ 的假設, 代表採用 random effect model 是合適的。#

15.20

(9)

```
Call:
lm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
    freelunch, data = star_data)

Residuals:
    Min       1Q   Median       3Q      Max
-107.220  -20.214   -3.935   14.339   185.956

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  437.76425    1.34622  325.180   < 2e-16 ***
small         5.82282     0.98933    5.886  4.19e-09 ***
aide          0.81784     0.95299    0.858    0.391
tchexper      0.49247     0.06956    7.080  1.61e-12 ***
boy          -6.15642     0.79613   -7.733  1.23e-14 ***
white_asian   3.90581     0.95361    4.096  4.26e-05 ***
freelunch    -14.77134     0.89025  -16.592   < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 5759 degrees of freedom
(因為不存在, 20 個觀察量被刪除了)
Multiple R-squared:  0.09685,    Adjusted R-squared:  0.09591
F-statistic: 102.9 on 6 and 5759 DF,  p-value: < 2.2e-16
```

小班制對學生閱讀分數有正向顯著效果,
助教對成績影響有限,
老師經驗與豐富可提升學生閱讀分數,
男生分數顯著低於女生,
亞裔白人分數顯著較高
并

(b)

```
Call:
lm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
  freelunch, data = p_star, model = "within")

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Residuals:
    Min.      1st Qu.        Median      3rd Qu.       Max.
-102.6381   -16.7834    -2.8473    12.7591    198.4169

Coefficients:
            Estimate Std. Error t-value Pr(>|t|)
small      6.490231    0.912962   7.1090 1.313e-12 ***
aide       0.996087    0.881693   1.1297  0.2586
tchexper   0.285567    0.070845   4.0309 5.629e-05 ***
boy       -5.455941    0.727589   -7.4987 7.440e-14 ***
white_asian 8.028019    1.535656   5.2277 1.777e-07 ***
freelunch -14.593572    0.880006  -16.5835 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 4628000
Residual Sum of Squares: 4268900
R-Squared: 0.077592
Adj. R-Squared: 0.063954
F-statistic: 79.6471 on 6 and 5681 DF, p-value: < 2.22e-16
```

加入學校 fixed effect 後, white_Asian 係數變化最大(3.91 → 8.03), 說明OLS低估種族效果, tchexper 的係數下降(0.492 → 0.286)說明老師經驗效果來自學校層面的差異。整體而言, 主要的結論尚未改變。
井

(c)

```
> #c小題
> #比較 FE 模型與 Pooled OLS 模型是否有系統性差異
> print(pFtest(model_fe, model_ols))
```

F test for individual effects

```
data: readscore ~ small + aide + tchexper + boy + white_asian + freelunch
F = 16.698, df1 = 78, df2 = 5681, p-value < 2.2e-16
alternative hypothesis: significant effects
```

$P\text{-value} = 2.2 \times 10^{-16} < 0.05$, 代表 school fixed effect 顯著, 應加入模型。

當解釋變數與學校效果不相關時, 固定效果對其他係數估計影響較小。
井

(d)

```
【題 d】學校隨機效應模型 (Random Effects) 估計結果
> print(summary(model_re))
Oneway (individual) effect Random Effect Model
(Swamy-Arora's transformation)

Call:
plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
  freelunch, data = p_star, model = "random")

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Effects:
            var std.dev share
idiosyncratic 751.43   27.41  0.829
individual    155.31   12.46  0.171
theta:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.6470  0.7225  0.7523  0.7541  0.7831  0.8153

Residuals:
    Min.      1st Qu.        Median      Mean      3rd Qu.       Max.
 -97.4830  -17.2361    -3.2822    12.8027  192.3460

Coefficients:
            Estimate Std. Error z-value Pr(>|z|)
(Intercept) 436.12674    2.064782 211.2217 < 2.2e-16 ***
small      6.458722    0.912548  7.0777 1.466e-12 ***
aide       0.992146    0.881159  1.1260  0.2602
tchexper   0.302679    0.070292  4.3060 3.582e-05 ***
boy       -5.512081    0.727639  -7.5753 3.581e-14 ***
white_asian 7.350477    1.431376  5.1533 2.818e-07 ***
freelunch -14.584332    0.874676 -16.6740 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 6158000
Residual Sum of Squares: 4332100
R-Squared: 0.29655
Adj. R-Squared: 0.29582
Chisq: 493.205 on 6 DF, p-value: < 2.22e-16
> cat("Breusch-Pagan LM 檢定結果 (H0: 無隨機效應)\n")
Breusch-Pagan LM 檢定結果 (H0: 無隨機效應)
> print(plmtest(model_re, type = "bp"))

Lagrange Multiplier Test - (Breusch-Pagan)

data: readscore ~ small + aide + tchexper + boy + white_asian + freelunch
chisq = 6677.4, df = 1, p-value < 2.2e-16
alternative hypothesis: significant effects
```

Part (a)

```
Call:
lm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
  freelunch, data = star_data)

Residuals:
    Min.      1Q      Median      3Q      Max
-107.220  -20.214   -3.935   14.339   185.956

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 437.76425    1.34622 325.180 < 2e-16 ***
small      6.48282    0.88931  7.386 4.19e-09 ***
aide       0.81784    0.95299  0.858  0.391
tchexper   0.49247    0.06956  7.080 1.61e-12 ***
boy       -6.15642    0.79613  -7.733 1.23e-14 ***
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freelunch -14.77134    0.89025 -16.592 < 2e-16 ***
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Residual standard error: 30.19 on 5759 degrees of freedom
(因不存在 20 個隨機變數而刪除)
Multiple R-squared: 0.09685, Adjusted R-squared: 0.09591
F-statistic: 102.9 on 6 and 5759 DF, p-value: < 2.2e-16
```

Part (b)

```
Call:
plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
  freelunch, data = p_star, model = "within")

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Residuals:
    Min.      1st Qu.        Median      3rd Qu.       Max.
-102.6381   -16.7834    -2.8473    12.7591    198.4169

Coefficients:
            Estimate Std. Error t-value Pr(>|t|)
small      6.490231    0.912962   7.1090 1.313e-12 ***
aide       0.996087    0.881693   1.1297  0.2586
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boy       -5.455941    0.727589   -7.4987 7.440e-14 ***
white_asian 8.028019    1.535656   5.2277 1.777e-07 ***
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Total Sum of Squares: 4628000
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R-Squared: 0.077592
Adj. R-Squared: 0.063954
F-statistic: 79.6471 on 6 and 5681 DF, p-value: < 2.22e-16
```

FE 和 RE 的數值非常接近, 表示 RE model 的模型偏誤沒有太嚴重, 但仍有些許差異。

white_asian 和 tchexper 是最容易與學校效果有相關。

4;

$LM(chisq) = 6677.4$, $P\text{-value} = 2.2 \times 10^{-16} < 0.05$ $\therefore$ reject $H_0 (a_1 = 0)$ at $\alpha = 0.05$

表示學校之間存在非隨機的異質性, 故不應使用 Pooled OLS。
井

(e)

```
> print(results, row.names = FALSE)
Variable      b_FE      b_RE    se_FE    se_RE    t_stat p_value
small      6.49023    6.45872  0.91296  0.91255    1.1460  0.2518
aide       0.99609    0.99215  0.88169  0.88116    0.1284  0.8978
tchexper   0.28557    0.30268  0.07084  0.07029   -1.9377  0.0527
boy      -5.45594   -5.51208  0.72759  0.72764         NA         NA
white_asian 8.02802    7.35048  1.53566  1.43138    1.2181  0.2232
freelunch -14.59357  -14.58433  0.88001  0.87468   -0.0956  0.9239
```

small, aide, tchexper, white_asian, freelunch 的 p-value 均超過 0.05, 說明使用 RE model 與使用 FE model 沒有顯著差異, 因此使用 RE model 即可。而因為 boy 變數的 $se(b_{FE}) < se(b_{RE})$, 故 $se(b_{FE})^2 - se(b_{RE})^2 < 0$, 此時無法對負值開根號, 故 boy 變數的 t 值與 p 值無法計算。

(f)

```
--- OLS 迴歸估計結果 (穩健標準誤) ---
> print(coeftest(mundlak, vcov = robust_vcov))

t test of coefficients:

            Estimate Std. Error   t value   Pr(>|t|)
(Intercept)  432.58566   15.07626  28.6932 < 2.2e-16 ***
small         6.57140    1.66298   3.9516 7.858e-05 ***
aide          1.07422    1.40159   0.7664 0.44345
tchexper      0.28967    0.13844   2.0925 0.03644 *
boy          -6.06843    0.74659  -8.1282 5.303e-16 ***
white_asian   3.50134    2.72318   1.2858 0.19858
freelunch    -15.13468    1.05431 -14.3550 < 2.2e-16 ***
mean_small   -21.67352   22.44946  -0.9654 0.33437
mean_aide     8.97460   20.23866   0.4434 0.65747
mean_tchexper 1.05851    0.56388   1.8772 0.06054 .
mean_freelunch 1.47474    5.61557   0.2626 0.79286
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Linear hypothesis test:

```
mean_small = 0
mean_aide = 0
mean_tchexper = 0
mean_freelunch = 0
```

Model 1: restricted model

Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelunch + mean_small + mean_aide + mean_tchexper + mean_freelunch

Note: Coefficient covariance matrix supplied.

```
Res.Df Df    F Pr(>F)
1    5695
2    5691  4 1.508 0.1969
```

加入的學校特質平均項均不顯著, 說明學校未獲見察到的個體效果與自變數之間沒有系統性的關聯, 因此可使用 RE model, 效率更高。